Md. Abdur Rahman

Dhaka, Bangladesh | moabdurrahmancs@gmail.com | Website | Google Scholar | GitHub | LinkedIn

Education

Military Institute of Science and Technology — *MSc in Computer Science and Engineering (ongoing)* *Apr 2026 – Present*

Southeast University — *BSc in Computer Science and Engineering* *Oct 2021 – Oct 2025*

- **CGPA:** 3.87/4.00 — **CGPA (Last 60 Credits):** 3.97/4.00
- **Relevant Coursework:** Artificial Intelligence, Image Processing, Data Mining, Research Methodology

Experience

Research Assistant — *Islamic University of Technology (IUT), OIC* *Gazipur, BD*
Nov 2025 – Present

- Engineered a Federated Learning framework for fintech data with a novel aggregation technique for non-IID settings.
- Pioneered hyperbolic Graph Neural Networks for blockchain fraud detection and causal inference methods for high-dimensional auditing.
- Authored five manuscripts accepted in Q1 journals on decentralized AI, graph-based fraud detection.

Research Assistant — *CAIR, Southeast University* *Dhaka, BD*
Jan 2025 – Oct 2025

- Designed hybrid transformer-based architectures for medical image segmentation and classification, fusing convolutional and attention mechanisms to improve diagnostic accuracy on MRI and X-ray datasets.
- Integrated explainable AI methods (SHAP, LIME) into clinical decision-support pipelines, enabling interpretable model outputs for high-stakes diagnostic settings.
- Authored five papers: three accepted at peer-reviewed conferences, two under review in Q1 journals on medical AI and interpretable deep learning.

Selected Publications

J *= Journal* ***C*** *= Conference* ***W*** *= Workshop*

[J.1] **MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum
Frontiers in Plant Science [Q1]

[J.2] **Integrating Domain-Specific Convolutional Features with Swin Transformers for Transparent Cervical Cancer Diagnostics** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Md Fokrul Islam KHAN, Md Zia Uddin, Ulysse Teller Masao Côté-Allard, Abdul Kadar Muhammad Masum
Frontiers in Oncology [Q2]

[J.3] **FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Public Health [Q1]

[J.4] **DFU-GCNet: A Global Context-Enhanced Inception Network for Robust and Interpretable Diabetic Foot Ulcer Classification** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Digital Health [Q1]

[J.5] **Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Ramona Birău, Virgil Popescu
IEEE Accessc [Q1]

[J.6] **Predicting Corporate Bankruptcy Through a Scalable and Interpretable Knowledge Distillation Framework** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Thanh Lam Nguyen, Hasan Dinçer, Serhat Yuksel
Discover Artificial Intelligence [Q1]

[J.7] **EcoStack-Pro: An Adaptive Federated Learning Framework for Interpretable ESG Auditing Across Heterogeneous Industrial Sectors** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Farzan M. Noori, Md Zia Uddin
Frontiers in Artificial Intelligence [Q1]

[J.8] **Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**
Applied Computational Intelligence and Soft Computing [Q1]

[J.9] **Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers** [Paper]
Rezwanul Huque Khan, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Faisal Md. Abdur Rahman, Abdul Kadar Muhammad Masum
Applied Computational Intelligence and Soft Computing [Q1]

[C.1] **FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification** [Paper]
Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay
International Conference on Data Science, AI and Applications (ICDSAIA 2025)

[C.2] **X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Mohammad Mahmudur Rahman, Iffat Sania Hossain, Abu Kowshir Bitto, Prof. Dr. Abdul Kadar Muhammad Masum
28th International Conference on Computer and Information Technology (ICCIT 2025)

[C.3] **DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution** [Paper]
Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam
7th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2025)

[W.1] **Do LLMs Distinguish Between Halal and Haram? Benchmarking Islamic Cultural Alignment in General vs. Arabic-Centric SLMs** [Paper]
Md Abdur Rahman, Md Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
6th Muslims in ML Workshop, ICML 2026

[W.2] **Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR** [Paper]
Sanjid Hasan, ***Md Abdur Rahman***
6th Muslims in ML Workshop, ICML 2026

[W.3] **SFL-MTKD++: Split ViT Federated Learning with Adaptive Multi-Teacher Distillation for Interpretable Malaria Diagnosis** [Paper]
Md Tofael Ahmed Bhuiyan, ***Md Abdur Rahman***, Abdul Kadar Muhammad Masum
6th Muslims in ML Workshop, ICML 2026

[W.4] **AlzFed-XAI: High-Fidelity Interpretable Alzheimer's Diagnosis with Privacy-Preserving Federated Learning** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
5th Muslims in ML Workshop, NeurIPS 2025

[W.5] **ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
2nd Bangla Language Processing (BLP) Workshop, AACL-IJCNLP 2025

[W.6] **BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Abdul Kadar Muhammad Masum
2nd Bangla Language Processing (BLP) Workshop, AACL-IJCNLP 2025

[W.7] **Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification** [Paper]
Md. Abdur Rahman, Md Ashiqur Rahman
4th Ukrainian Natural Language Processing Workshop (UNLP), ACL 2025

Under Review Publications

1. **HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation** [1st Author]
   Submitted to *IEEE Open Journal of the Computer Society*
2. **BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention** [1st Author]
   Submitted to *PLOS ONE*
3. **HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation** [1st Author]
   Submitted to *Engineering Reports*
4. **Rerank-RRG: Cross-Modal Phrase Reranking for Retrieval-Augmented Radiology Report Generation** [2nd Author]
   Submitted to *29th International Conference on Computer and Information Technology (ICCIT)*

Experience with Research Grant

Assisted in proposal preparation for the following research grants:

- **Closing the Employability Gap: A Machine Learning–Driven Framework to Predict and Improve Student Placement Outcomes at University** *August 2026*
  Co-PI, IRT Research Grant, Southeast University
  Funded amount: $4,000
- **Bridging the Gap: A Predictive Analytics Framework for Enhancing Engineering Student Employability and Placement Outcomes in Bangladesh** *September 2026*
  Team Member, Bangladesh Accreditation Council (BAC) Research Grant
  Funded amount: $3,000

Honors & Awards

- **3rd Position (Poster Presentation)**, Student Research Day 2025 [Certificate]
- **Champion**, Fragments of Feeling: A Contextual Emotion Challenge (2025) [GitHub]
- **1st Runners-up**, CalorieQuest: Predicting Energy Burn with Data (2025) [GitHub]
- **2nd Runners-up**, DU AI Challenge (2025) [GitHub]
- **2nd Runners-up**, REACT 2026 Datathon, hosted by IEEE SEU Student Branch [GitHub]
- **Finalist (Top 10)**, CUET ETE Datathon 2025 [GitHub]

Academic Service

- **Program Committee**, 21st BEA Workshop co-located with **ACL 2026**
- **Reviewer**, 7th Muslims in ML Workshop at **NeurIPS 2026**
- **Reviewer**, Frontiers in Artificial Intelligence
- **Reviewer**, Frontiers in Medicine
- **Reviewer**, Green Technologies and Sustainability

Extracurricular Activities

Research Executive (ExCom), IEEE Computer Society Bangladesh Chapter Graduate Student Member Forum *2026*

Sub-Executive (Research & Development), IEEE Computer Society Southeast University Student Branch Chapter *2025*

Skills

Languages: C, C++, Java, Python
Frameworks: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot
Technologies: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Git/GitHub, Cloud Computing, LaTeX

Professional Certification

- **AWS Certified Cloud Practitioner**, Amazon Web Services *Dec 2025 – Dec 2028*
- **GitHub Foundations Certification**, GitHub *Nov 2024 – Nov 2027*